Doctorate Research Project

Extreme heat and its impact in hospitalizations and mortality in Rio de Janeiro: risk levels and sociospatial disparities

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Co-advisor abroad: **Rachel Lowe**

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Epidemiology in Public Health Doctorate Program
Sergio Arouca National School of Public Health (ENSP/Fiocruz)
March 2026

CIE
CENTRO DE INTELIGÊNCIA
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- Objectives
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Context

- Known direct and indirect effects of extreme heat in human health[1,2]
  - Affects known morbidity and mortality patterns

1. BELL, Michelle L.; GASPARRINI, Antonio; BENJAMIN, Georges C. Climate Change, Extreme Heat, and Health. New England Journal of Medicine, v. 390, n. 19, p. 1793–1801, 2024.
2. SORENSEN, Cecilia; HESS, Jeremy. Treatment and Prevention of Heat-Related Illness. New England Journal of Medicine, v. 387, n. 15, p. 1404–1413, 12 out. 2022.

Context

- Known direct and indirect effects of extreme heat in human health[1,2]
  - Affects known morbidity and mortality patterns
- Scaling frequency and intensity of such events across the globe, including Latin America[1,3-5]

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3. PERKINS, S. E.; ALEXANDER, L. V.; NAIRN, J. R. Increasing frequency, intensity and duration of observed global heatwaves and warm spells. Geophysical Research Letters, v. 39, n. 20, 2012.
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5. BELL, Michelle L. et al. Vulnerability to heat-related mortality in Latin America: a case-crossover study in São Paulo, Brazil, Santiago, Chile and Mexico City, Mexico. International Journal of Epidemiology, v. 37, n. 4, p. 796–804, 1 ago. 2008.

Context

- Elderly, women, children and certain occupational groups known to be at higher risk[1,2]

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- Elderly, women, children and certain occupational groups known to be at higher risk[1,2]
- Evidence on social and racial disparities still controversial[6,7]

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6. GRONLUND, Carina J. Racial and Socioeconomic Disparities in Heat-Related Health Effects and Their Mechanisms: a Review. Current Epidemiology Reports, v. 1, n. 3, p. 165–173, 1 set. 2014.
7. BAKHTSIYARAVA, Maryia et al. Modification of temperature-related human mortality by area-level socioeconomic and demographic characteristics in Latin American cities. Social Science & Medicine, v. 317, p. 115526, 1 jan. 2023.

Context

- Elderly, women, children and certain occupational groups known to be at higher risk[1,2]
- Evidence on social and racial disparities still controversial[6,7]
- Spatially, challenges exist regarding data granularity and methods that incorporate a spatial component[8]

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8. QUIJAL-ZAMORANO, Marcos. Temperature impacts on mortality at different spatial and temporal scales. Ph.D. Thesis—[S.l.]: Universitat Pompeu Fabra, 14 nov. 2024.

Context

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- Remarkable 2023 heat wave

Taylor Swift fan died of heat exhaustion at Rio concert, forensics report finds

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https://www.theguardian.com/music/2023/dec/28/taylor-swift-fan-cause-of-death-rio-brazil-heat-exhaustion-ana-clara-benevides

Context

- Rio de Janeiro: latin-american city known for high temperatures (“Rio 40°C”)
- Remarkable 2023 heat wave
  - Rio launches first Brazilian heat protocol in 2024

Taylor Swift fan died of heat exhaustion at Rio concert, forensics report finds

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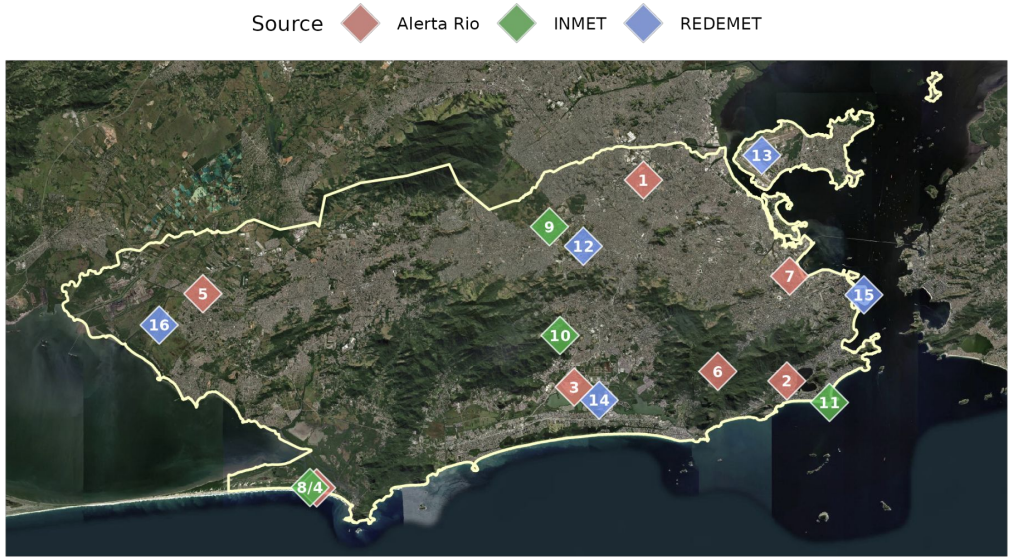
Objectives

- To identify causes of medical visits, hospitalizations and mortality associated with extreme heat, measured through different indices in Rio de Janeiro, as well as social and spatial disparities in the distribution of such outcomes.

Climate data

- Hourly **temperature (T)** and **humidity (H)** measurements from 16 weather stations in the city of Rio, from 3 different sources



1. IRAJA; 2. JARDIM_BOTANICO; 3. BARRA_RIOCENTRO; 4. GUARATIBA; 5. SANTA_CRUZ; 6. ALTO_DA_BOA_VISTA; 7. SAO_CRISTOVAO; 8. MARAMBAIA; 9. VILA MILITAR; 10. RIO DE JANEIRO - JACAREPAGUA; 11. FORTE DE COPACABANA; 12. BASE AEREA DOS AFONSOS; 13. AEROPORTO INTERNACIONAL DO RIO DE JANEIRO; 14. AEROPORTO DE JACAREPAGUA; 15. AEROPORTO SANTOS DUMONT; 16. BASE AEREA DE SANTA CRUZ

Outcome data

- Mortality data
  - All deaths for Rio residents, by cause of death
  - Geolocalization of such deaths
  - Systematic collection

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Cause of death group	ICD-10 codes
Influenza and Pneumonia	J09-J18
Chronic lower respiratory diseases	J40-J47
Other diseases of the respiratory system	J00–J06, J20–J39, J60–J70, J80–J86, J90–J96, J97–J99, R09.2, U04
Hypertensive diseases	I10-I15
Ischemic heart disease	I20-I25
Heart failure	I50
Cerebrovascular diseases	I60-I69
Other disease of the circulatory system	I00–I09, I26–I49, I51, I52, I70–I99
Malignant neoplasms	C00-C97
Alzheimer disease and dementia	G30, G31, F01, F03
Diabetes	E10-E14
Renal Failure	N17-N19
Sepsis	A40-A41

Table 1. Included ICD-10 codes of underlying causes of death, possibly heat-related.

Outcome data

- Hospitalization data
  - All hospitalizations in public health facilities
  - Possibility to look at intermediary outcomes

Outcome data

- Primary care and emergency care visits
- Eletronic health record data

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239 PRIMARY AND SECONDARY CARE UNITS

46 HOSPITALS AND EMERGENCY CARE FACILITIES

38 MENTAL HEALTH CENTERS

BRAZIL HAS THE "SUS"

•Unified Health System, a universal and public healthcare system.

Outcome data

- Primary care and emergency care visits
- Eletronic health record data
- Much more unstable data source

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Data Analysis

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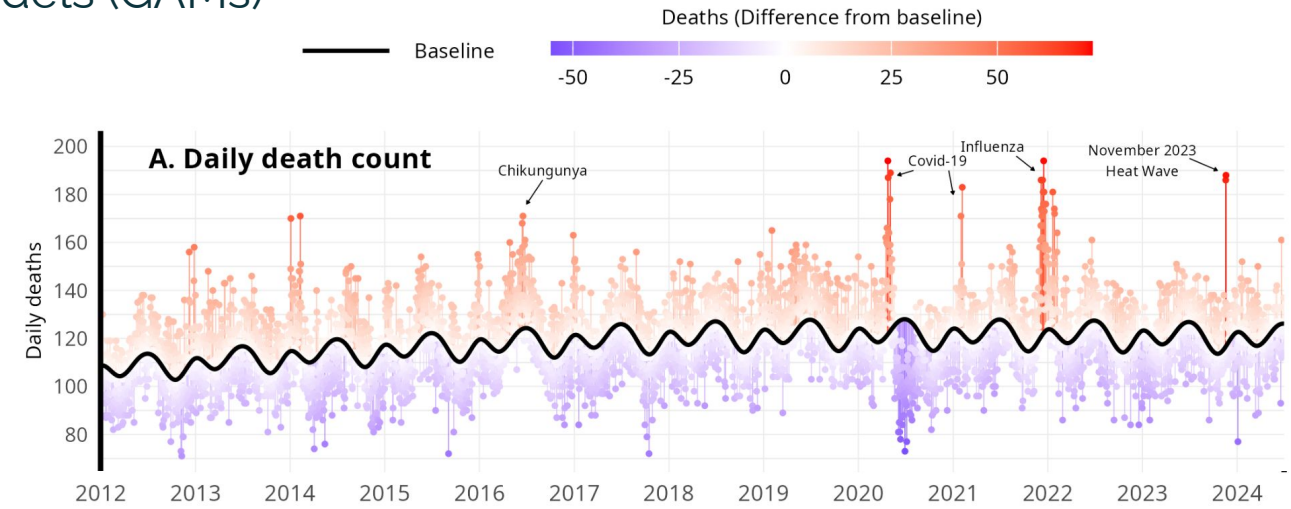

Figure: Baseline for daily deaths (solid line) among **all selected causes**, and difference between observed and baseline

Data Analysis

- Exposure-response curves obtained through Distributed Lag Non-Linear Models (DLNMs)
  - Up to 10 days lag
  - “U-shaped” curves expected

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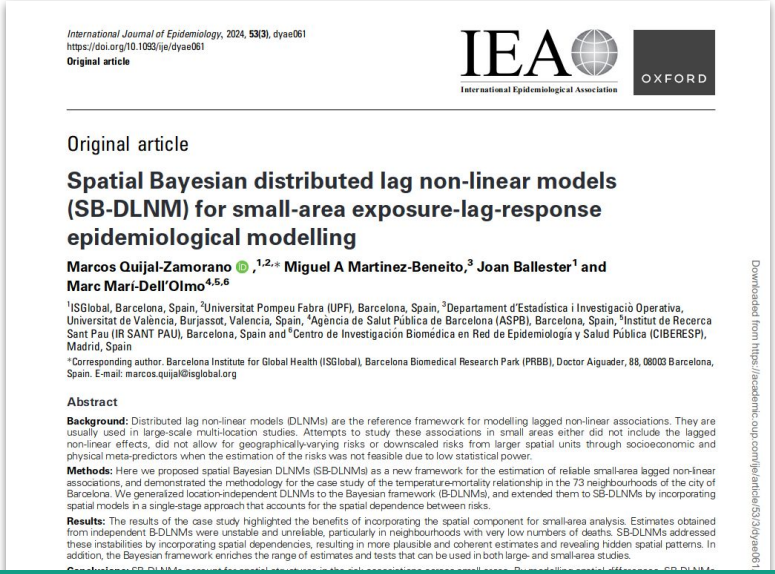
- Exposure-response curves obtained through Distributed Lag Non-Linear Models (DLNMs)
  - Up to 10 days lag
  - “U-shaped” curves expected
- Models for different exposure metrics
  - Average temperature, average heat index, exposure period

Data Analysis

- For spatialization of outcome: Spatial-Bayesian DLNMs?



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https://doi.org/10.1093/ije/dyae061
Original article

IEA International Epidemiological Association

OXFORD

Original article

Spatial Bayesian distributed lag non-linear models (SB-DLNM) for small-area exposure-lag-response epidemiological modelling

Marcos Quijal-Zamorano[1,2,*], **Miguel A Martinez-Beneito**,[3] **Joan Ballester**[1] **and Marc Marí-Dell'Olmo**[4,5,6]

[1]ISGlobal, Barcelona, Spain, [2]Universitat Pompeu Fabra (UPF), Barcelona, Spain, [3]Departament d'Estadística i Investigació Operativa, Universitat de València, Burjassot, Valencia, Spain, [4]Agència de Salut Pública de Barcelona (ASPB), Barcelona, Spain, [5]Institut de Recerca Sant Pau (IR SANT PAU), Barcelona, Spain and [6]Centro de Investigación Biomédica en Red de Epidemiología y Salud Pública (CIBERESP), Madrid, Spain

*Corresponding author. Barcelona Institute for Global Health (ISGlobal), Barcelona Biomedical Research Park (PRBB), Doctor Aiguader, 88, 08003 Barcelona, Spain. E-mail: marcos.quijal@isglobal.org

Abstract

Background: Distributed lag non-linear models (DLNMs) are the reference framework for modelling lagged non-linear associations. They are usually used in large-scale multi-location studies. Attempts to study these associations in small areas either did not include the lagged non-linear effects, did not allow for geographically-varying risks or downscaled risks from larger spatial units through socioeconomic and physical meta-predictors when the estimation of the risks was not feasible due to low statistical power.

Methods: Here we proposed spatial Bayesian DLNMs (SB-DLNMs) as a new framework for the estimation of reliable small-area lagged non-linear associations, and demonstrated the methodology for the case study of the temperature-mortality relationship in the 73 neighbourhoods of the city of Barcelona. We generalized location-independent DLNMs to the Bayesian framework (B-DLNMs), and extended them to SB-DLNMs by incorporating spatial models in a single-stage approach that accounts for the spatial dependence between risks.

Results: The results of the case study highlighted the benefits of incorporating the spatial component for small-area analysis. Estimates obtained from independent B-DLNMs were unstable and unreliable, particularly in neighbourhoods with very low numbers of deaths. SB-DLNMs addressed these instabilities by incorporating spatial dependencies, resulting in more plausible and coherent estimates and revealing hidden spatial patterns. In addition, the Bayesian framework enriches the range of estimates and tests that can be used in both large- and small-area studies.

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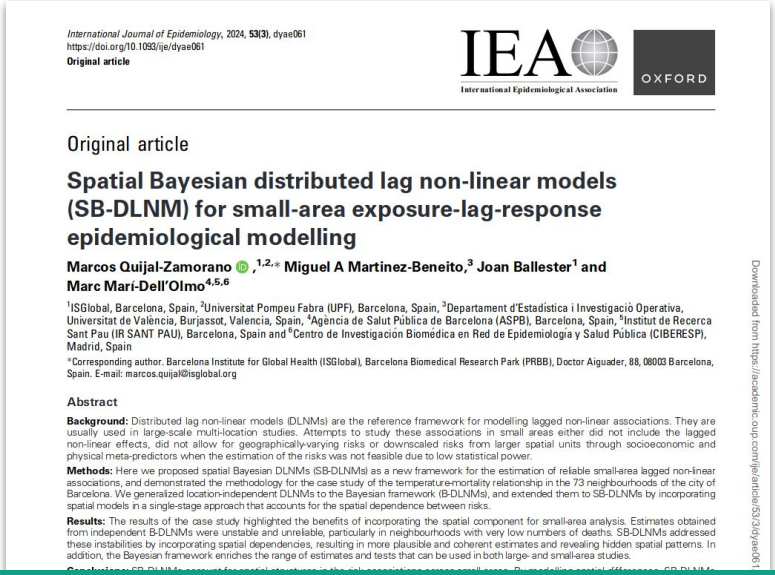

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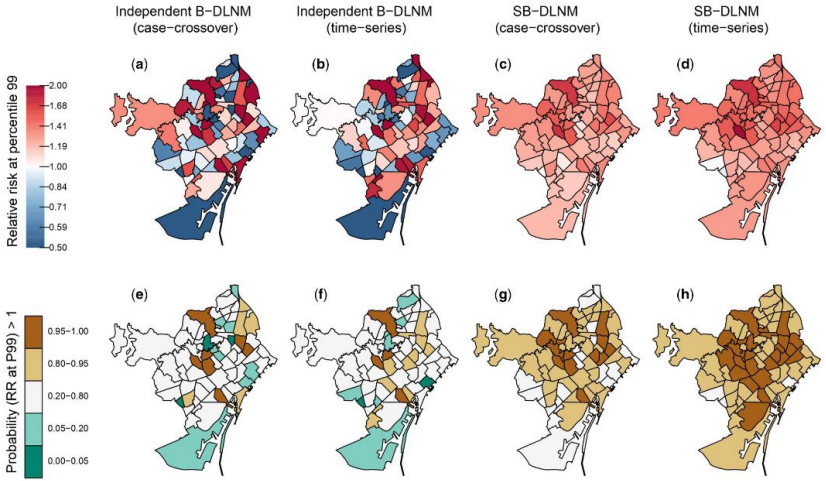

Extracted from: QUIJAL-ZAMORANO, Marcos et al. Spatial Bayesian distributed lag non-linear models (SB-DLNM) for small-area exposure-lag-response epidemiological modelling. International Journal of Epidemiology, v. 53, n. 3, p. dyae061, 11 abr. 2024.

Results (so far)

- Preprint (submitted to Environmental Epidemiology) containing temporal analysis of mortality data

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Quantifying heat exposure and its related mortality in Rio de Janeiro City: evidence to support Rio's recent heat protocol

João Henrique de Araujo Morais, Débora Medeiros de Oliveira e Cruz, Valeria Saraceni, Caroline Dias Ferreira, Gislani Mateus Oliveira Aguilar, Oswaldo Gonçalves Cruz

doi: https://doi.org/10.1101/2025.01.17.25320740

This article is a preprint and has not been peer-reviewed [what does this mean?]. It reports new medical research that has yet to be evaluated and so should *not* be used to guide clinical practice.

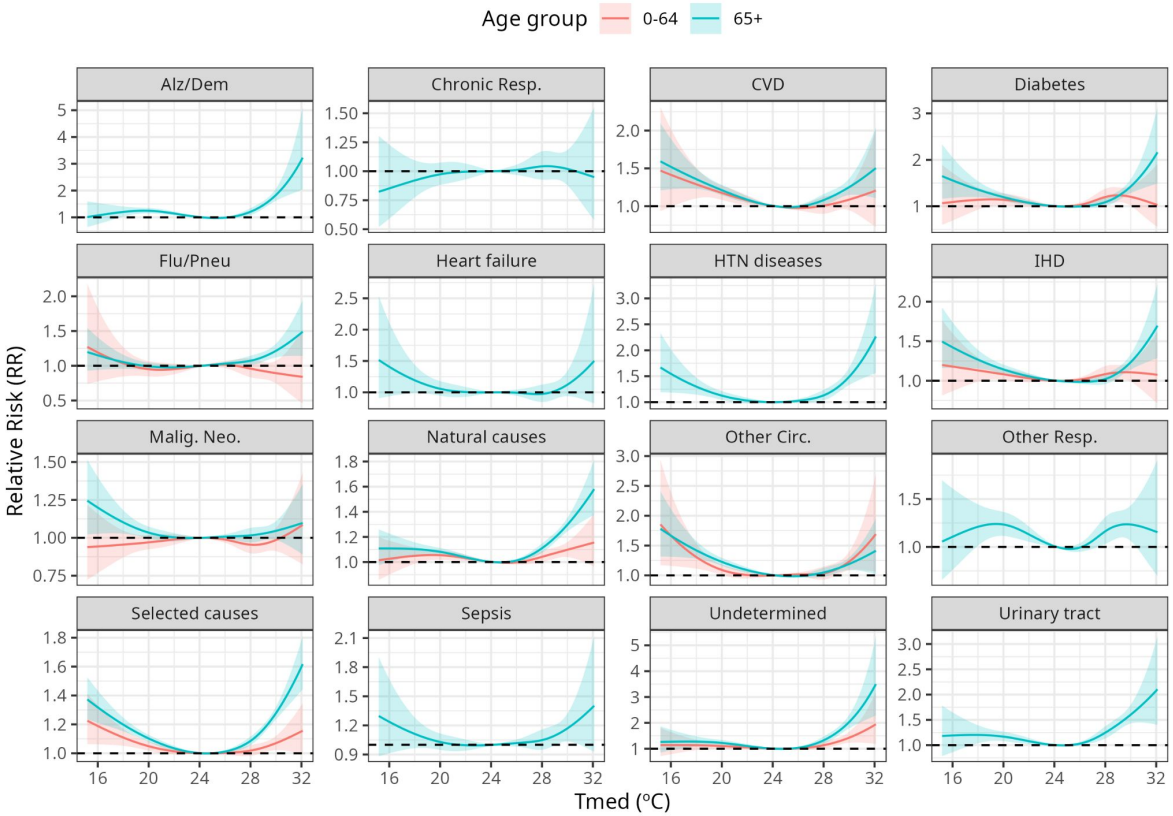
Abstract | **Full Text** | Info/History | Metrics | Preview PDF

Abstract

Under a climate change scenario, extreme heat episodes show an increase in frequency and intensity, with a scaling impact in Latin American cities. Recently, Rio de Janeiro City developed its heat protocol, using the amount of hours spent over specific heat thresholds as its trigger metric. This study gathers mortality data by 17 different causes in Rio, in a 12.5 year period (2012-2024). We use Distributed Lag Non-Linear Models (DLNM) to assess the relationship between different heat exposure metrics to mortality among the young (< 65) and elderly

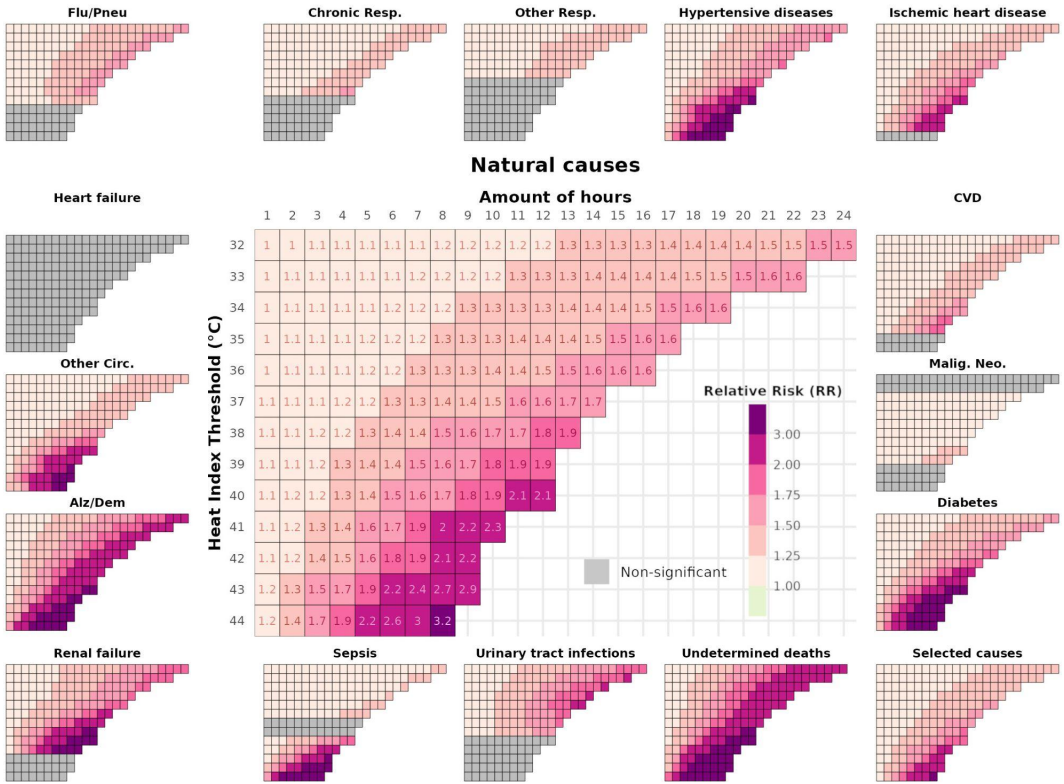
Results (so far)

- Exposure-response curves by age of group and cause of death



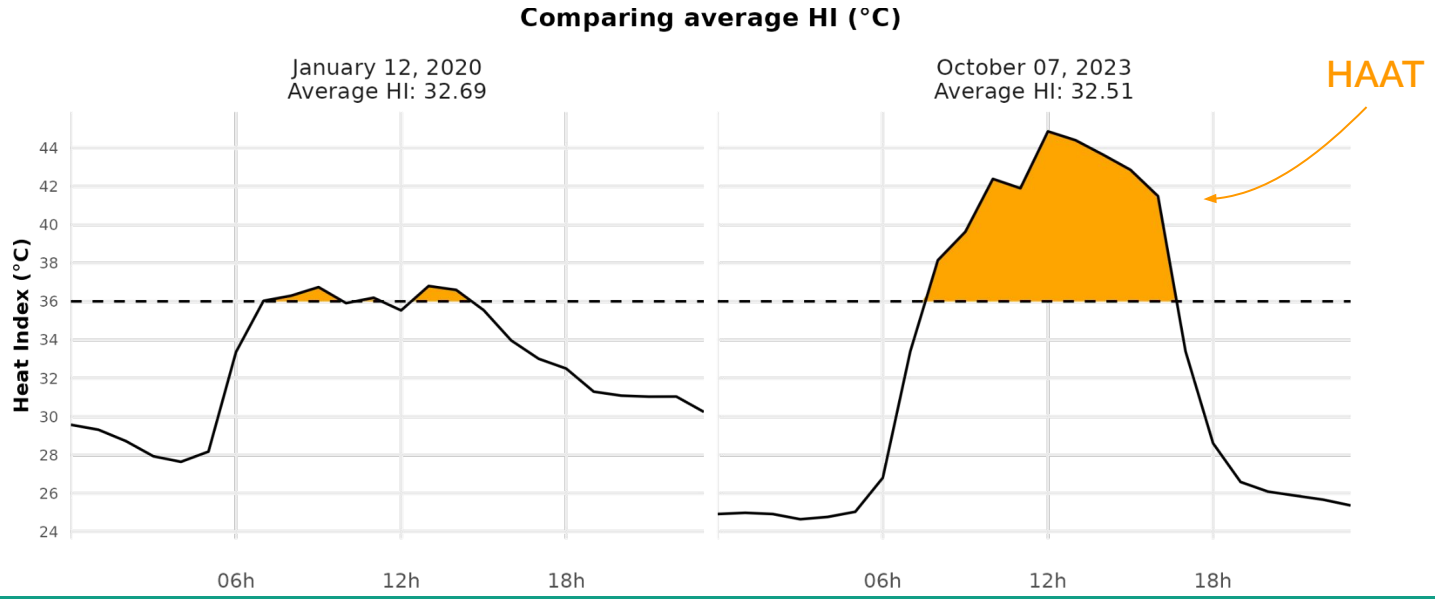
Results (so far)

- Effects by number of hours spent above heat thresholds



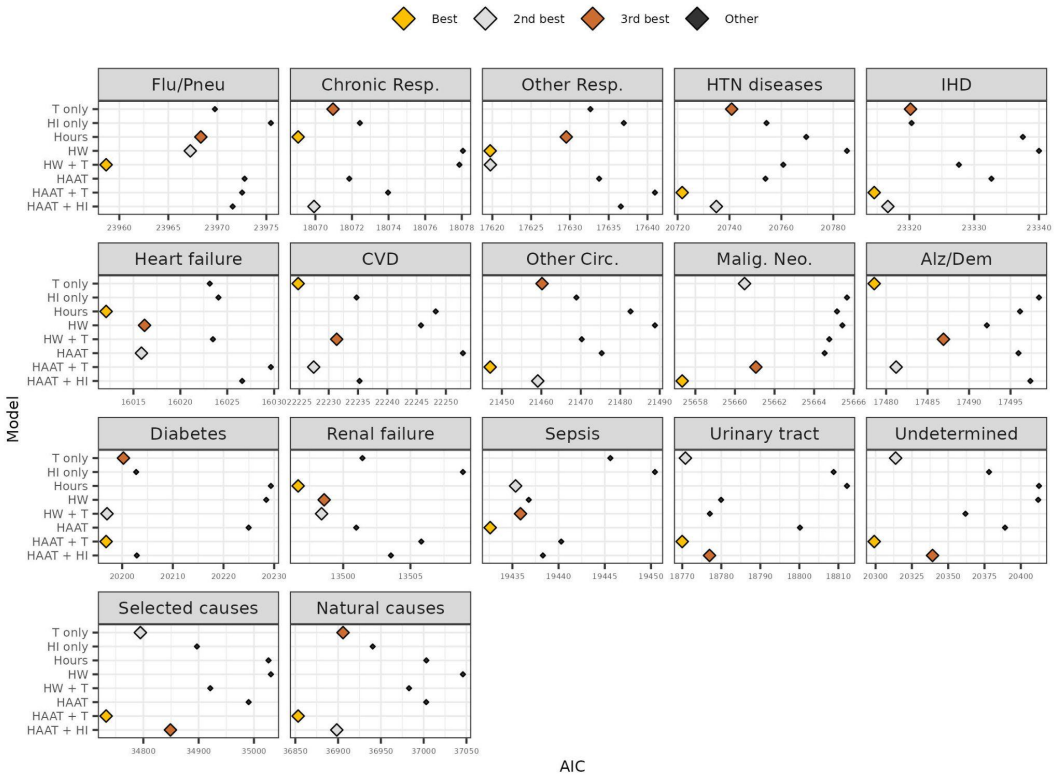
Results (so far)

- Proposal of a heat exposure metric: Heat Area Above a Threshold (HAAT)



Results (so far)

- Exposure metrics comparison
- Heat Index (HI) with no relevant benefit over Temperature (T)
- HAAT + T performing best for 10/17 causes



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- Outcome can be spatialized (georreferenced mortality data points), what about exposure?
- How to incorporate such components in a DLNM model?

Thank you!

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